

Multiple Choice (10 marks)

1. Calculate the Larmor frequency for a proton in a magnetic field of 1 T.
 - (a) 23.56 GHz
 - (b) 50 MHz
 - (c) 100 MHz
 - (d) 42.58 MHz
 - (e) 1 GHz
2. Calculate the normality of a solution 40 ml of which is required to neutralize 0.56 g of KOH.
 - (a) 0.5 N
 - (b) 0.2 N
 - (c) 0.75 N
 - (d) 0.1 N
 - (e) 1 N
3. Each resonance form of the nitrate ion, NO_3^- , has how many sigma and how many pi bonds?
 - (a) 4 sigma and 1 pi
 - (b) 3 sigma and 1 pi
 - (c) 0 sigma and 4 pi
 - (d) 3 sigma and 2 pi
 - (e) 2 sigma and 1 pi
4. In a chemical reaction requiring two atoms of phosphorus for five atoms of oxygen, how many grams of oxygen are required by 3.10 g of phosphorus?
 - (a) 9.30 g O
 - (b) 2.50 g O
 - (c) 4.00 g O
 - (d) 5.00 g O
 - (e) 3.10 g O
5. A sample of 255mg of neon occupies 3.00dm^3 at 122K. Use the perfect gas law to calculate the pressure of the gas.

- (a) 4.20×10^{-2} atm
- (b) 5.55×10^{-2} atm
- (c) 6.33×10^{-2} atm
- (d) 3.15×10^{-2} atm
- (e) 8.21×10^{-2} atm

True / False (10 marks)

Answer True or False

6. True or False: The magnetogyric ratio of ^{23}Na is $20.78 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$ based on its NMR frequency in a 750 MHz spectrometer.
7. True or False: When a solid melts, it experiences a positive enthalpy change and a positive entropy change due to increased molecular disorder.
8. True or False: The molar weight of the non-electrolyte in solution is approximately 100 g solute per mole solute.
9. True or False: Hydrogen bonds are stronger than dipole-dipole interactions, which are in turn stronger than induced dipole forces.
10. True or False: The energy of one mole of UV photons with a wavelength of 300 nm is approximately 450 kJ/mol.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of electrolysis in an electrolytic cell using a zinc salt solution, and calculate the equivalent weight of zinc based on the deposition of 3.27 g of zinc from a current of 5.36 A over 30 minutes.
12. Analyze the electrochemical cell $\text{Pt}/\text{H}/\text{HCl} (0.2 \text{ M})/\text{HCl} (0.8 \text{ M})/\text{H}/\text{Pt}$ at 25°C with $E = 0.83 \text{ V}$. Calculate the cell voltage and discuss the factors influencing your result.

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